Modeling the Deposition Process of Silver Nano-Islands

February 9, 2026

1 Introduction

The morphological and optical evolution of thin metallic films during deposition has long been a subject of both fundamental interest and technological relevance. When metals are deposited on weakly interacting substrates, they typically grow through a sequence of nucleation, island formation, coalescence, and eventually percolation, rather than forming uniform layers directly. These dynamic processes give rise to nano-island morphologies whose dielectric response evolves non-linearly with deposition time, critically impacting the plasmonic and optoelectronic properties of the resulting films.

A substantial body of research has focused on the deposition dynamics themselves. In situ scattering studies, most prominently the work of Schwartzkopf et al. [1], revealed distinct growth regimes during sputter deposition of gold clusters: nucleation, diffusion-mediated coalescence, adsorption-driven growth, and percolation. The signatures of these transitions appear in both the peak position and FWHM of the scattering features, as well as in Yoneda bands, reflecting the complex interplay of diffusion, aggregation, and vertical growth. Kinetic Monte Carlo and analytical models have reinforced this picture: Gervilla et al. [2] demonstrated the role of facet migration in coalescence, while Lu and co-workers identified conditions for coalescence-controlled versus coalescence-free regimes under pulsed flux deposition [3, 4]. The broader phenomenology of thin-film morphological evolution on weakly interacting substrates has been comprehensively reviewed by Sarakinos [5].

Our approach in this report is not to reproduce these dynamics from first principles, but instead to construct a simplified, qualitative framework—a toy model—that captures their main features. Following the phenomenology observed by Schwartzkopf et al., we emulate the experimentally reported broadening–narrowing sequence of the island size distribution by prescribing a lognormal dispersion with a time-dependent geometric spread $\sigma_{geo}(s)$, modeled as a Gaussian-like bump centered around the coalescence window. This ansatz reproduces the mid-process broadening and late-stage narrowing trends while preserving a nearly constant filling fraction.

In parallel, the optical properties arising from the distribution of metallic islands in air require a dedicated description. Here we couple the morphological toy model with the effective medium framework of Battie et al. [6], who extended the Maxwell–Garnett–Mie formulation to account for nanoparticle size dispersion. This allows us to link each deposition step, as represented in our simplified dynamic model, with the corresponding effective dielectric function. In this way, we bridge the qualitative kinetics of deposition with the quantitative optical response.

Building on these elements, the present work introduces a stepwise deposition model for silver nano-island films. By explicitly parametrizing thickness growth, the evolution of size dispersion, and the filling fraction, and by coupling these to effective medium optics, our framework provides a compact yet physically motivated route to reproduce spectroscopic trends observed during deposition and to interpret non-monotonic behaviors in terms of realistic, experimentally inspired growth scenarios.

2 Deposition Steps and Thickness Growth

We describe the deposition as a sequence of N measurement steps. The nominal film thickness after the full process is denoted d_∞. Each step d_s ($s = 1, \ldots, N$) corresponds to a probe of the system at a finite deposition time.

In the simplest case of linear growth, the thickness evolves as

$$d(s) = d_\infty \frac{s}{N}, \tag{1}$$

so that the last step recovers the total thickness $d(N) = d_\infty$.

To reflect experimental practice, where the first measured point corresponds to a non-zero deposition time (e.g., 10 s), we generalize by introducing an explicit first thickness d_1. The interpolated growth law becomes

$$d(s) = d_1 + (d_\infty - d_1)\frac{s-1}{N-1}, \tag{2}$$

which ensures $d(1) = d_1$ and $d(N) = d_\infty$.

3 Island Size Distribution

At each deposition step the film morphology is represented by an average island radius R_{ave} and a standard deviation σ_{nm}, assuming a lognormal size distribution consistent with nanoparticle growth kinetics [6]. The cumulative thickness is linked to these parameters through

$$d = R_{\text{ave}} + \alpha\, \sigma_{nm}, \tag{3}$$

with α a proportionality constant.

The dispersion is defined as

$$k = \frac{\sigma_{nm}}{R_{\text{ave}}} = \exp[(\ln \sigma_{geo})^2] - 1, \tag{4}$$

where σ_{geo} is the geometric width parameter of the lognormal distribution. Solving gives

$$R_{\text{ave}} = \frac{d}{1+\alpha k}, \qquad \sigma_{nm} = R_{\text{ave}}\, k. \tag{5}$$

4 Evolution of the Geometric Width

The evolution of $\sigma_{geo}(s)$ with deposition step captures the transition from sparse nucleation to coalescence and broadening of the size distribution. We explored three representative laws:

1. Constant: $\sigma_{geo}(s) = \sigma_{min}$,
2. Linear interpolation: from σ_{min} to σ_{peak} across N steps,
3. Gaussian broadening:

$$\sigma_{geo}(s) = \sigma_{min} + (\sigma_{peak} - \sigma_{min}) \exp\left[-\frac{(s-\mu)^2}{2w^2}\right], \tag{6}$$

with center μ (default $N/2$) and width w (default $N/4$).

5 Filling Fraction of Metallic Inclusions

The filling fraction $f(s)$ describes the volumetric ratio of silver in the composite medium. From geometric scaling arguments,

$$f(s) \propto \frac{1}{1+\alpha k(s)}. \tag{7}$$

To anchor this to the experimental reference at the first step, we normalize as

$$f(s) = f_0 \frac{1+\alpha k(1)}{1+\alpha k(s)}, \tag{8}$$

with f_0 the measured filling fraction at $s = 1$.

To further mimic reduced packing efficiency for broad distributions, we apply an empirical correction:

$$f(s) \to \frac{f(s)}{1+\beta[\sigma_{geo}(s)-1]^2}, \tag{9}$$

with β an adjustable parameter ($\beta = 0$ recovers the uncorrected case).

6 Optical Constants

For the host medium (air) we used a dielectric constant $\varepsilon_2 = 1.0006$. The silver permittivity $\varepsilon_1(\omega)$ was obtained by interpolating ellipsometric measurements performed on a homogeneous silver slab deposited under the same conditions. This procedure ensures consistency between the morphological growth model and the experimentally determined optical constants.

7 Results and Discussion

The deposition model was implemented in a Bash script (`sweep_ds.bash`), which sweeps over deposition steps, computes the morphological parameters, and outputs the corresponding effective medium parameters. Example results obtained with parameters $f_0 = 0.25$, $\sigma_{min} = 1.6$, $\sigma_{peak} = 1.8$, and $d_1 = 2.5$ yielded close agreement with the available experimental dataset.

Figure 1 illustrates the evolution of the effective permittivity during the simulated deposition process. Panels (a) and (b) display the spectral dependence of the real and imaginary parts of the effective permittivity, respectively, for successive deposition steps. The characteristic resonant features shift as the nano-islands evolve, consistent with progressive changes in the filling fraction and size dispersion.

Panels (c) and (d) track the evolution of characteristic spectral markers: the wavelength of maximum $\mathrm{Im}(\varepsilon)$ and the two epsilon-near-zero (ENZ) crossings. Up to step 3, these quantities follow a smooth trend. However, beyond step 3, a reversed step-like behavior emerges, particularly in the ENZ positions. This apparent anomaly originates from the non-monotonic growth of the size dispersion parameter $\sigma_{geo}(s)$, which was modeled using a Gaussian broadening law.

The adoption of a Gaussian ansatz for $\sigma_{geo}(s)$ was motivated by experimental reports of a broadening–narrowing sequence during cluster growth. Such non-monotonic behavior has been observed in kinetic Monte Carlo simulations and in real-time scattering experiments on sputter-deposited gold clusters by Schwartzkopf et al. [**?**]. In their in situ mGISAXS study, distinct transitions between nucleation, coalescence, and adsorption-driven growth were identified, leading to a non-linear evolution of the cluster size distribution. Our phenomenological choice of a Gaussian-like law for $\sigma_{geo}(s)$ captures this behavior in a compact form, ensuring that the model remains transparent about its empirical nature while being firmly anchored in experimental evidence.

Thus, the results highlight how the interplay between morphological dispersion and filling fraction drives the effective medium response. The reversed step-like trend after step 3

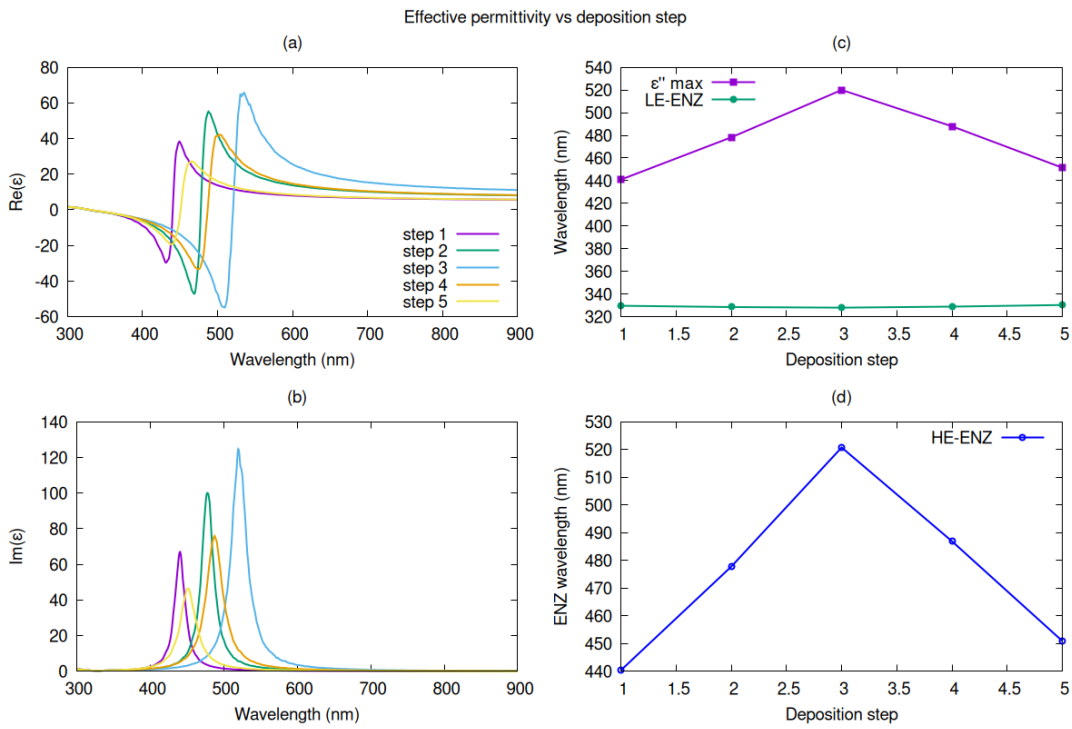

Figure 1: Simulation of effective permittivity during silver nano-island deposition. Curves correspond to deposition steps (step 1 ... step 5). Parameters: final thickness 10.0 nm, first-step thickness 2.5 nm, $\alpha = 1.0$, $f_0 = 0.25$, sigma-law=gauss, $\sigma_{\text{min}} = 1.6$, $\sigma_{\text{peak}} = 1.8$. Detailed step-by-step values are reported in the `summary.dat` file. Panels: (a) Real part of the effective permittivity (Re(ε)) versus wavelength for each deposition step. (b) Imaginary part of the effective permittivity (Im(ε)). (c) Evolution of the wavelength corresponding to the maximum of Im(ε) (red points) and the low-energy ENZ crossing (blue points) as a function of deposition step. (d) Evolution of the high-energy ENZ crossing as a function of deposition step.

is not an artifact, but rather a manifestation of the complex, non-monotonic evolution of island morphology during deposition. This emphasizes the importance of incorporating realistic, literature-motivated parametrizations of $\sigma_{geo}(s)$ in order to reproduce experimental features faithfully.

8 Conclusion

We have developed a flexible, physics-motivated deposition model that captures the morphological evolution of silver nano-islands during sputter deposition. By tuning d_1, the $\sigma_{geo}(s)$ law, and the packing correction β, the model reproduces key features of the experimental growth curves, including the transition from nucleation to coalescence. This approach provides a robust starting point for integrating spectroscopic data and refining the microscopic interpretation of deposition dynamics.

References

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